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Energy storage apparatus

Abstract

An energy storage apparatus includes an outer case that accommodates an energy storage device and a reinforcing member. The reinforcing member includes the pair of reinforcing walls disposed so as to sandwich the two wall portions of the outer case from an outside of the outer case, the two wall portions oppositely facing each other, and the connecting portion connecting the pair of reinforcing walls to each other.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6228528	12/2000	Burkholder et al.	N/A	N/A
2009/0197154	12/2008	Takasaki et al.	N/A	N/A
2011/0294000	12/2010	Kim et al.	N/A	N/A
2014/0205878	12/2013	Ohgitani et al.	N/A	N/A
2016/0329538	12/2015	Hughes	N/A	N/A
2018/0111501	12/2017	Takeuchi et al.	N/A	N/A
2019/0140232	12/2018	Ogawa et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
S62-077864	12/1986	JP	N/A
H08-169242	12/1995	JP	N/A
2009-087757	12/2008	JP	N/A
2011-249315	12/2010	JP	N/A
2012-160347	12/2011	JP	N/A
2013-133044	12/2012	JP	N/A
2014-139895	12/2013	JP	N/A
2014-170678	12/2013	JP	N/A
2018-029014	12/2017	JP	N/A
WO2016/186139	12/2017	JP	N/A
2018-056097	12/2017	JP	N/A
2018-065513	12/2017	JP	N/A
2018-521447	12/2017	JP	N/A
2018-195378	12/2017	JP	N/A
2019-087526	12/2018	JP	N/A

OTHER PUBLICATIONS

Machine Translation JP 2018029014 (Year: 2018). cited by examiner

International Search Report (ISR) (PCT Form PCT/ISA/210), in PCT/JP2021/013166, dated Jun. 8, 2021. cited by applicant

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to an energy storage device, and an energy storage apparatus including an outer case that accommodates the energy storage device.

BACKGROUND ART

(2) Patent Document 1 discloses an energy storage apparatus that includes: a body portion that accommodates a battery unit that has a plurality of batteries; and a reinforcing structure that is mounted on the body portion. This reinforcing structure includes: support struts that project upward beyond the body portion; and a receiving member that is fixed to upper portions of the support struts. Accordingly, even when any heavy object falls on the energy storage apparatus, a load (an impact load) can be received by the reinforcing structure and hence, the deformation, the rupture, and the like of the body portion are prevented.

PRIOR ART DOCUMENT

Patent Document

(3) Patent Document 1: WO 2016/186139 A

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(4) In the above-mentioned conventional energy storage apparatus, in preparation for a case where some object impinges on the energy storage apparatus, the reinforcing structure (a reinforcing member) is disposed on the body portion (an outer case). However, a force acts on the outer case of the energy storage apparatus not only from the outside of the outer case but also from the inside of the outer case. When a gas is discharged from the energy storage device in the outer case, a pressure in the outer case (an internal pressure) sharply rises so that the outer case receives a force (the internal pressure) from the inside. Accordingly, the outer case is deformed so as to expand outward, and as a result, there is a possibility that a damage such as cracking occurs in the outer case. When a damage occurs on the outer case, there is a possibility that an unsafe phenomenon occurs. For example, a phenomenon that a gas in the outer case is leaked through a damaged portion occurs. Therefore, how to suppress such a defect of the outer case caused by a force that the outer case receives from the inside of the outer case is a crucial task from a viewpoint of increasing a safety of the energy storage apparatus.

(5) The inventors of the present invention have made the present invention by focusing on the above-mentioned task from a novel viewpoint, and it is an object of the present invention to provide an energy storage apparatus where the safety of the energy storage apparatus is enhanced.

Means for Solving the Problems

(6) An energy storage apparatus according to an aspect of the present invention includes: an outer case that accommodates an energy storage device; and a reinforcing member including a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions oppositely facing each other, and a connecting portion that connects the pair of reinforcing walls to each other, wherein rigidity of each of the pair of reinforcing walls is higher than rigidity of the connecting portion.

(7) An energy storage apparatus according to another aspect of the present invention includes: an outer case that accommodates an energy storage device; and a reinforcing member including a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions oppositely facing each other, and a connecting portion that connects the pair of reinforcing walls to each other, wherein the reinforcing member includes a first

reinforcing member and a second reinforcing member, each of the first reinforcing member and the second reinforcing member includes two plate-shaped portions disposed opposite to each other and the connecting portion connecting the two plate-shaped portions to each other, and each of the pair of reinforcing walls is formed by making one of the two plate-shaped portions of the first reinforcing member and one of the two plate-shaped portions of the second reinforcing member overlap with each other.

(8) An energy storage apparatus according to another aspect of the present invention includes: an outer case that accommodates an energy storage device; and a reinforcing member including a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions opposedly facing each other, and a connecting portion that connects the pair of reinforcing walls to each other, wherein a thickness of each of the pair of reinforcing walls is larger than a thickness of the connecting portion.

Advantages of the Invention

(9) According to the present invention, it is possible to provide an energy storage apparatus where the safety of the energy storage apparatus is enhanced.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating an external appearance of an energy storage apparatus according to an embodiment.

(2) FIG. 2 is an exploded perspective view of an energy storage apparatus according to the embodiment.

(3) FIG. 3 is a perspective view illustrating the configuration of a reinforcing member and the periphery thereof according to the embodiment.

(4) FIG. 4 is an exploded perspective view of the reinforcing member according to the embodiment.

(5) FIG. 5 is a first cross-sectional view illustrating a structural relationship between the reinforcing member and an outer case according to the embodiment.

(6) FIG. 6 is a second cross-sectional view illustrating a structural relationship between the reinforcing member and the outer case according to the embodiment.

MODE FOR CARRYING OUT THE INVENTION

(7) An energy storage apparatus according to an aspect of the present invention includes: an outer case that accommodates an energy storage device; and a reinforcing member including a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions opposedly facing each other, and a connecting portion that connects the pair of reinforcing walls to each other, wherein rigidity of each of the pair of reinforcing walls is higher than rigidity of the connecting portion.

(8) According to this configuration, the pair of reinforcing walls is connected to each other so as to restrict the movement of the pair of reinforcing walls to the outside. Accordingly, when a wall portion of the outer case receives an external force from the inside, the expansion of the wall portion to the outside is suppressed. As a result, the occurrence of a drawback such as a damage on the outer case is suppressed. Such an advantageous effect can be obtained by the reinforcing member that is mounted on the outer case from the outside and hence, the advantageous effect described above can be acquired without changing the designing of the outer case. Furthermore, the reinforcing wall has relatively high rigidity and hence, the expansion of the wall portion of the outer case can be more reliably suppressed. Further, with respect to the connecting portion where a demand for high rigidity is not so high compared to the reinforcing wall, a thickness of the connecting portion can be reduced or the connecting portion can be made thin. That is, it is possible

to effectively suppress deformation of the outer case (the expansion or bulging of the wall portion) while realizing the reduction of weight, downsizing or the like of the reinforcing member. As described above, according to the energy storage apparatus of this aspect, the safety of the energy storage apparatus can be enhanced.

(9) The pair of reinforcing walls may be connected by the connecting portion disposed on each of both sides in a width direction that intersects with an oppositely facing direction of the two wall portions.

(10) According to this configuration, the pair of reinforcing walls is connected to each other on both sides in the width direction and hence, the pair of reinforcing walls can exert a high restraint force on the outer case. Accordingly, an effect of suppressing the expansion of the outer case is increased, and this contributes to enhancement of safety of the energy storage apparatus.

(11) The energy storage device includes a case that includes a pair of long-side surfaces oppositely facing each other, and a plurality of the energy storage devices that are arranged side by side such that the pair of long-side surfaces faces the oppositely facing direction of the two wall portions may be accommodated in the outer case.

(12) In a case where a plurality of energy storage devices are accommodated in the outer case in a state where the plurality of energy storage devices are arranged side by side, when the energy storage devices each have a prismatic shape, in general, the energy storage devices are arranged such that long-side surfaces of the energy storage devices face the arrangement direction of the energy storage devices. The long-side surfaces of the prismatic energy storage device are likely to expand and hence, an array of the energy storage devices formed of the plurality of energy storage devices is likely to expand in the arrangement direction of the plurality of energy storage devices. Accordingly, it is safe to say that the two opposing wall portions are likely to expand outward by receiving an expanding force applied to two wall portions. In this regard, in the energy storage apparatus of this aspect, the reinforcing member is disposed in a posture suitable for restraining the two wall portions and hence, the expansion of the outer case can be effectively suppressed. That is, the reinforcing member is disposed so as to press the portion of the outer case which is likely to expand and hence, the safety of the energy storage apparatus is enhanced.

(13) The reinforcing member includes a first reinforcing member and a second reinforcing member, each of the first reinforcing member and the second reinforcing member includes two plate-shaped portions disposed opposite to each other and the connecting portion connecting the two plate-shaped portions to each other, and each of the pair of reinforcing walls may be formed by making one of the two plate-shaped portions of the first reinforcing member and one of the two plate-shaped portions of the second reinforcing member overlap with each other.

(14) According to this configuration, the reinforcing member can be configured by combining two members (first and second reinforcing members) having the same shape with each other. Accordingly, it is possible to obtain a reinforcing wall having high rigidity that is formed of two plate-shaped portions. By combining the two members such that the connecting portions of the two members face each other, it is also possible to obtain a reinforcing member which can suppress expansion of the outer case in a well-balanced manner. Accordingly, it is possible to obtain a reinforcing member which efficiently suppresses the expansion of the outer case with a simple configuration, and this contributes to the enhancement of the safety of the energy storage apparatus.

(15) An energy storage apparatus according to another aspect of the present invention includes: an outer case that accommodates an energy storage device; and a reinforcing member including a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions oppositely facing each other, and a connecting portion that connects the pair of reinforcing walls to each other, wherein the reinforcing member includes a first reinforcing member and a second reinforcing member, each of the first reinforcing member and the second reinforcing member includes two plate-shaped portions disposed opposite to each other and the connecting portion connecting the two plate-shaped portions to each other, and each of the pair

of reinforcing walls is formed by making one of the two plate-shaped portions of the first reinforcing member and one of the two plate-shaped portions of the second reinforcing member overlap with each other.

(16) According to this configuration, the pair of reinforcing walls is connected to each other so as to restrict the movement of the pair of reinforcing walls to the outside and hence, the expansion of the wall portion of the outer case to the outside is suppressed. As a result, the occurrence of a damage or the like on the outer case is suppressed. The reinforcing member can be configured by combining two members (first and second reinforcing members) having the same shape with each other. Accordingly, it is possible to obtain a reinforcing wall having high rigidity that is formed of two plate-shaped portions. By combining the two members such that the connecting portions of the two members face each other, it is also possible to obtain a reinforcing member which can suppress expansion of the outer case in a well-balanced manner. Accordingly, it is possible to obtain a reinforcing member which efficiently suppresses the expansion of the outer case with a simple configuration. As described above, according to the energy storage apparatus of this aspect, the safety of the energy storage apparatus can be enhanced.

(17) An energy storage apparatus according to another aspect of the present invention includes: an outer case that accommodates an energy storage device; and a reinforcing member including a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions oppositely facing each other, and a connecting portion that connects the pair of reinforcing walls to each other, wherein a thickness of each of the pair of reinforcing walls is larger than a thickness of the connecting portion.

(18) According to this configuration, the pair of reinforcing walls is connected to each other so as to restrict the movement of the pair of reinforcing walls to the outside. Accordingly, when a wall portion of the outer case receives an external force from the inside, the expansion of the wall portion to the outside is suppressed. As a result, the occurrence of a drawback such as a damage on the outer case is suppressed. Such an advantageous effect can be obtained by the reinforcing member that is mounted on the outer case from the outside and hence, the advantageous effect described above can be acquired without changing the designing of the outer case. Further, the reinforcing wall has relatively large thickness and hence, the expansion of the wall portion of the outer case can be more reliably suppressed. Further, with respect to the connecting portion where a demand for large thickness is not so high compared to the reinforcing wall, the connecting portion can be formed with a relatively thin wall thickness. That is, it is possible to effectively suppress deformation of the outer case (the expansion or bulging of the wall portion) while realizing the reduction of weight, downsizing or the like of the reinforcing member. As described above, according to the energy storage apparatus of this aspect, the safety of the energy storage apparatus can be enhanced.

(19) Hereinafter, the energy storage apparatus according to the embodiment (including modifications of the embodiment) of the present invention is described with reference to the drawings. All embodiments described hereinafter are a comprehensive example or a specific example. In the following embodiment, numerical values, shapes, materials, constituent elements, arrangement positions and connection modes of the constituent elements, and the like are provided as examples, and are not intended to limit the present invention. In the respective drawings, sizes and the like are not strictly illustrated.

(20) In the following description and drawings, an arrangement direction along which a plurality of energy storage devices are arranged, a direction in which a pair of long-side surfaces of a case of an energy storage device are opposite to each other, or a thickness direction of the case is defined as an X-axis direction. A direction along which electrode terminals of one energy storage device are arranged or a direction in which a pair of short-side surfaces of the case of the energy storage device are opposite to each other is defined as a Y-axis direction. An arrangement direction of a body portion and a lid body in an outer case of the energy storage apparatus, an arrangement

direction of the energy storage devices and bus bars, or a vertical direction is defined as a Z axis direction. These X-axis direction, Y-axis direction, and Z-axis direction are directions that intersect with each other (orthogonal to each other in the embodiment and the modifications of the embodiment described later). A case is also considered where the Z-axis direction is not the vertical direction depending on a use mode. However, in the description made hereinafter, for the sake of convenience of the description, the description is made by assuming the Z-axis direction as the vertical direction.

(21) In the embodiments described later, in some cases, the expressions indicating the relative directions or the relative postures such as parallel or orthogonal are used. However, these expressions also include a case where constituent elements of the energy storage apparatus do not take such directions or postures in strict meaning of the terms. A state where two directions are parallel to each other means not only a state where these two directions are completely parallel to each other but also a state where these two directions are substantially parallel to each other, that is, a state where these two directions are parallel to each other with slight difference of about a few percent. In the description made hereinafter, an X-axis direction positive side indicates an arrow direction side of the X-axis, and an X-axis direction negative side indicates a side opposite to the X-axis direction positive side. The same goes for the Y-axis direction and the Z-axis direction.

EMBODIMENT

(22) [1. Description of Overall Configuration of Energy Storage Apparatus]

(23) First, the overall configuration of an energy storage apparatus **1** according to the embodiment will be described with reference to FIG. **1** and FIG. **2**. FIG. **1** is a perspective view illustrating an external appearance of the energy storage apparatus **1** according to the embodiment. FIG. **2** is an exploded perspective view of the energy storage apparatus **1** according to the embodiment.

(24) The energy storage apparatus **1** is an apparatus into which electricity can be charged from the outside and from which electricity can be discharged to the outside. In this embodiment, the energy storage apparatus **1** has an approximately rectangular parallelepiped shape. The energy storage apparatus **1** may be a battery module (assembled battery) used in an electricity storage application, a power source application, or the like. To be more specific, the energy storage apparatus **1** may be used as a battery or the like for driving a mobile body such as an automobile, a motorcycle, a watercraft, a ship, a snowmobile, an agriculture machine, a construction machine, or a railway vehicle for an electric railway, or may be used as a battery for starting an engine of the mobile body. As the above-described automobile, an electric vehicle (EV), a hybrid electric vehicle (HEV), a plug-in hybrid electric vehicle (PHEV), and a gasoline automobile are exemplified. As the above-described railway vehicle for an electric railway, an electric train, a monorail, and a linear motor car are exemplified. The energy storage apparatus **1** can also be used as a stationary battery or the like used as a home-use battery, a generator, or the like.

(25) As illustrated in FIG. **1** and FIG. **2**, the energy storage apparatus **1** includes a plurality of energy storage devices **20**, an outer case **10** that accommodates the plurality of energy storage devices **20**, and a reinforcing member **100** mounted on the outer case **10**. In this embodiment, eight energy storage devices **20** are accommodated in the outer case **10**. The number of energy storage devices **20** included in the energy storage apparatus **1** is not limited to 8. It is sufficient for the energy storage apparatus **1** to include one or more energy storage devices **20**. In this embodiment, one energy storage device array **24** is formed of the plurality of energy storage devices **20** arranged side by side in the X axis direction. The energy storage device array **24** may include a spacer, an insulating film, and the like not illustrated in the drawings.

(26) The outer case **10** includes a body portion **12** which accommodates the energy storage device array **24** and a lid body **11**, and a bus bar plate **17** is disposed between the energy storage device array **24** accommodated in the body portion **12** and the lid body **11**. A plurality of bus bars **33** are held on the bus bar plate **17**, and the plurality of bus bars **33** are covered with bus bar covers **70** and **75**. A connection unit **80** including a control circuit and the like is disposed between the bus bar

plate **17** and the lid body **11**.

(27) The outer case **10** is a case (module case) having a rectangular shape (box shape) that forms an outer profile of the energy storage apparatus **1**. That is, the outer case **10** is a member that fixes the energy storage device array **24**, the bus bar plate **17**, and the like at predetermined positions, and protects these constitutional elements from an impact or the like. For example, the outer case **10** is made of an insulating member such as polycarbonate (PC), polypropylene (PP), polyethylene (PE), polystyrene (PS), a polyphenylene sulfide resin (PPS), polyphenylene ether (PPE (including modified PPE)), polyethylene terephthalate (PET), polybutylene terephthalate (PBT), polyether ether ketone (PEEK), tetrafluoroethylene/perfluoroalkyl vinyl ether (PFA), polytetrafluoroethylene (PTFE), polyether sulfone (PES), an ABS resin, or a composite material of the above-mentioned materials, or metal to which insulation coating is applied.

(28) The lid body **11** included in the outer case **10** is a rectangular member which closes an opening portion **15** of the body portion **12**, and includes an external terminal **91** on a positive electrode side and an external terminal **92** on a negative electrode side. The external terminals **91** and **92** are electrically connected to the plurality of energy storage devices **20** via the connection unit **80** and the bus bars **33**. The energy storage apparatus **1** charges electricity from the outside and discharges electricity to the outside via the external terminals **91**, **92**. The external terminals **91**, **92** are respectively formed of a conductive member made of metal such as aluminum, or an aluminum alloy.

(29) The lid body **11** is provided with a ventilation chamber (not illustrated in the drawing) through which gas moving from one of the inside and the outside of the outer case **10** to the other passes, and an exhaust tube **90** that communicates the inside of the ventilation chamber with the outside of the outer case **10**. The gas inside the outer case **10** is discharged to the outside of the outer case **10** through the ventilation chamber and the exhaust tube **90**. More specifically, in the ventilation chamber, a valve member that is opened when a pressure (internal pressure) inside the outer case **10** increases to a predetermined value is disposed. Accordingly, in a normal state, even when foreign matter such as water or dust flows into the ventilation chamber through the exhaust tube **90**, the inflow of the foreign matter into the inside of the outer case **10** is substantially prevented by the valve member. When the gas is discharged from the energy storage device **20** and the internal pressure of the outer case **10** becomes equal to or higher than the predetermined value, the valve member is brought into an open state, and the gas inside the outer case **10** is discharged from the exhaust tube **90** to the outside of the outer case **10** through the ventilation chamber.

(30) The body portion **12** of the outer case **10** is a bottomed rectangular cylindrical housing (housing) in which the opening portion **15** for accommodating the energy storage device array **24** is formed. In a state where the opening portion **15** is closed by the lid body **11**, a peripheral edge of the opening portion **15** and the lid body **11** are joined to each other by thermal welding. Accordingly, airtightness in the opening portion **15** is secured.

(31) The energy storage device **20** is a secondary battery (battery cell) that can charge electricity and discharge electricity. More specifically, the energy storage device **20** is a nonaqueous electrolyte secondary battery such as a lithium ion secondary battery. The energy storage device **20** has a flat rectangular parallelepiped shape (prismatic shape). In this embodiment, eight pieces of energy storage devices **20** are arranged side by side in the X-axis direction as described above.

(32) In this embodiment, the energy storage device **20** includes the case **21** made of metal. The case **21** is a prismatic case having a pair of long-side surfaces **21a** facing each other and a pair of short-side surfaces **21b** facing each other. An electrode assembly, a current collector, an electrolyte solution, and the like are accommodated in the case **21**. In this embodiment, the plurality of energy storage devices **20** are arranged in a row in the X axis direction in a posture where the long-side surfaces **21a** are directed in the X axis direction (a posture where the short-side surfaces **21b** are parallel to the X axis direction).

(33) Electrode terminals **22** (a positive electrode terminal and a negative electrode terminal) made

of metal and electrically connected to the electrode assembly in the inside of the case **21** are provided on the lid plate **21c** of the case **21**. The lid plate **21c** of the case **21** is further provided with a gas release valve **23** for releasing the gas inside the case **21** to the outside. The gas release valve **23** has a function of releasing the gas inside the case **21** to the outside of the case **21** by releasing an internal pressure (by opening the valve) when the internal pressure of the case **21** increases due to vaporization of the electrolyte solution inside the case **21**. The gas release valve **23** having such a function is provided to each of the plurality of energy storage devices **20**. In this embodiment, as illustrated in FIG. 2, each of the plurality of energy storage devices **20** is arranged in a posture where the gas release valve **23** faces the Z-axis direction positive side.

(34) The energy storage device **20** is not limited to a nonaqueous electrolyte secondary battery. The energy storage device **20** may be a secondary battery other than the non-aqueous electrolyte secondary battery, or may be a capacitor. The energy storage device **20** may be a primary battery that allows a user to use stored electricity even when the user does not charge the energy storage device **20**. In this embodiment, the energy storage device **20** having the rectangular parallelepiped shape (prismatic shape) is illustrated. However, the shape of the energy storage device **20** is not limited to the rectangular parallelepiped shape. The energy storage device **20** may have a polygonal columnar shape other than the rectangular parallelepiped shape, a circular columnar shape, an elliptical columnar shape, or the like. Further, a laminate-type energy storage device may be provided to the energy storage apparatus **1** as the energy storage device **20**.

(35) The bus bar **33** is a rectangular plate-like member which is disposed on at least two energy storage devices **20** in a state where the bus bar **33** is held by the bus bar plate **17**, and the bus bar **33** electrically connects the electrode terminals **22** of at least these two energy storage devices **20** to each other. A material for forming the bus bar **33** is not particularly limited, and the bus bar **33** may be formed of metal such as aluminum, an aluminum alloy, copper, a copper alloy, or stainless steel or a combination thereof, or a conductive member other than metal. In the embodiment, using five bus bars **33**, four sets of energy storage device groups are formed where two energy storage devices **20** are connected with each other in parallel in each energy storage device group. Then, four sets of energy storage device groups are connected in series. The mode of electrical connection of the eight energy storage devices **20** is not particularly limited, and all of the eight energy storage devices **20** may be connected in series by seven bus bars.

(36) The connection unit **80** is a unit that includes a plurality of bus bars, a control board, and the like, and electrically connects the energy storage device array **24** and the external terminals **91** and **92** to each other. The control board that the connection unit **80** includes has a plurality of electric components. Using this plurality of electric components, a detection circuit that detects a state of each energy storage device **20**, a control circuit that controls charging and discharging of the energy storage devices **20**, and the like are formed. In this embodiment, the connection unit **80** is fixed to the bus bar plate **17**.

(37) The bus bar plate **17** is a resin-made member that holds the bus bars **33**. In this embodiment, the bus bar plate **17** is a member that holds the plurality of bus bars **33**, the connection unit **80**, and other wirings and the like (not illustrated in the drawing), and can perform positional restriction and the like of these members. A plurality of bus bar opening portions **17a** are formed in the bus bar plate **17**. The plurality of bus bar opening portions **17a** are provided for respectively holding the plurality of bus bars **33**. Apart of each of the plurality of bus bars **33** is exposed to a side where the plurality of energy storage devices **20** are disposed through the corresponding bus bar opening portion **17a**.

(38) In the middle of the bus bar plate **17** in the Y axis direction, a path forming portion **19** extending in the X axis direction and protruding toward the Z axis direction positive side is provided along the arrangement of the gas release valves **23** of the plurality of energy storage devices **20**. The path forming portion **19** covers all of the gas release valves **23** from the Z-axis direction positive side. As illustrated in FIG. 2, a path outlet **18** is provided at end portions of the

path forming portion **19** in a longitudinal direction of the path forming portion **19**, on both sides of an X-axis direction positive side and an X-axis direction negative side. Accordingly, the gas released from the energy storage device **20** mainly passes through the path outlet **18** and is discharged to the outside of the outer case **10** through the above-mentioned ventilation chamber and the above-mentioned exhaust tube **90**. The bus bar plate **17** configured as described above is fixed to the body portion **12** of the outer case **10** by a predetermined method such as adhesion or thermal welding.

(39) Each of the bus bar covers **70** and **75** is a resin-made member covering the plurality of bus bars **33** from above, and plays a role of electrically insulating the plurality of bus bars **33** and the connection unit **80** from each other.

(40) The reinforcing member **100** is a member which reinforces the outer case **10**. In the present embodiment, the reinforcing member **100** is disposed so as to surround the outer case **10**, and when the outer case **10** intends to expand due to an increase of an internal pressure, the expansion can be suppressed by the reinforcing member **100**. As illustrated in FIG. 1 and FIG. 2, a plurality of restricting members **200** are fixed to the reinforcing member **100** according to the present embodiment, and these restricting members **200** mainly have a function of pressing the lid body **11**. Each of the plurality of restricting members **200** is fixed to the reinforcing member **100** by welding or fastening with a bolt or a rivet. In the present embodiment, three restricting members **200** are fixed to the reinforcing member **100**, and in order to distinguish these three restricting members **200**, reference signs (**210**, **220**, **230**) different from each other are given to the three restricting members **200** as illustrated in FIG. 2. In the description made hereinafter, the matter described for the “restricting member **200**” is applied to each of the restricting members **210**, **220**, and **230**. Hereinafter, the configuration of the reinforcing member **100** and the periphery thereof will be further described with reference to FIG. 3 to FIG. 6.

(41) [2. Configuration of Reinforcing Member and Periphery Thereof]

(42) FIG. 3 is a perspective view illustrating the configuration of the reinforcing member **100** and the periphery thereof according to the embodiment. FIG. 3 illustrates a state where the reinforcing member **100** and spacers **151** and **152** are separated from the outer case **10**. FIG. 4 is an exploded perspective view of the reinforcing member **100** according to the embodiment. FIG. 5 is a first cross-sectional view illustrating a structural relationship between the reinforcing member **100** and the outer case **10** according to the embodiment. In FIG. 5, a transverse cross section (a cross section parallel to an XY plane) at a middle portion in the vertical direction (Z-axis direction) of the energy storage apparatus **1** is simply illustrated, and each energy storage device **20** is illustrated not in a cross-sectional view but in a plan view. The illustration of the restricting member **200** and the spacers **151** and **152** is omitted. FIG. 6 is a second cross-sectional view illustrating a structural relationship between the reinforcing member **100** and the outer case **10** according to the embodiment. FIG. 6 illustrates an end portion on an X-axis direction negative side in a VI-VI cross section in FIG. 5.

(43) As illustrated in FIG. 3, the reinforcing member **100** is an annular member as viewed in a plan view (when viewed from a Z-axis direction positive side) and is formed along four wall portions **13** of the outer case **10**. Specifically, the reinforcing member **100** includes a pair of reinforcing walls **101** disposed so as to sandwich the two wall portions **13** facing each other from the outside, and a connecting portion **102** connecting the pair of reinforcing walls **101** to each other. In the present embodiment, as illustrated in FIG. 3, the four wall portions **13** that the outer case **10** includes are distinguished into a pair of wall portions **13a** forming the short-side surfaces of the outer case **10** and a pair of wall portions **13b** forming the long-side surfaces of the outer case **10**. In this case, it can be expressed that the pair of reinforcing walls **101** is disposed so as to sandwich two wall portions **13a** facing each other from the outside.

(44) The reinforcing member **100** is a member made of metal such as iron or an aluminum alloy, and as illustrated in FIG. 3, is fixed to the outer case **10** in a state where the spacers **151** and **152**

are interposed between the reinforcing member **100** and the outer case **10**. The spacers **151** and **152** are members made of mica or a resin, and are members which protect the outer case **10** from the reinforcing member **100** having higher rigidity than the outer case **10** and electrically insulate the outer case **10** and the reinforcing member **100** from each other. As such a member, a mica molded article, or a resin having an electrical insulation property such as PP, PC, or PE is adopted in the same manner as the outer case **10**. The material for forming the spacers **151** and **152** such as a resin may contain a material for improving strength, durability, heat resistance, or the like, such as glass fiber.

(45) In the present embodiment, as illustrated in FIG. 4 and FIG. 5, the reinforcing member **100** is configured by combining a first reinforcing member **110** and a second reinforcing member **120**. Specifically, each of the two reinforcing walls **101** of the reinforcing member **100** is configured by overlapping a plate-shaped portion **111** of the first reinforcing member **110** and a plate-shaped portion **121** of the second reinforcing member **120**. One of the two connecting portions **102** of the reinforcing member **100** is a portion connecting the two plate-shaped portions **111** of the first reinforcing member **110**, and the other is a portion connecting the two plate-shaped portions **111** of the second reinforcing member **120**. In this manner, the reinforcing member **100** formed of the first reinforcing member **110** and the second reinforcing member **120** is fixed to the outer case **10** by the following procedure.

(46) First, the first reinforcing member **110** and the second reinforcing member **120** are disposed at positions illustrated in FIG. 5 with respect to the outer case **10**. Thereafter, the first reinforcing member **110** and the second reinforcing member **120** are pressurized from both sides in the Y axis direction, and the plate-shaped portion **111** and the plate-shaped portion **121** overlapped with each other in the X axis direction are joined to each other by spot welding or the like while maintaining such a pressed state. Accordingly, the outer case **10** is brought into a state where the outer case **10** is appropriately fastened by the reinforcing member **100**, and the reinforcing member **100** is brought into a state where the reinforcing member **100** is substantially fixed to the outer case **10**.

(47) Such an attachment operation of the reinforcing member **100** is performed in a state where the energy storage device array **24**, the connection unit **80** and the like are accommodated in the outer case **10** and the lid body **11** is joined to the opening portion **15** of the body portion **12**. At the time of attachment operation of the reinforcing member **100**, positions of the first reinforcing member **110** and the second reinforcing member **120** in the vertical direction (Z-axis direction) can be positioned by ribs **13e** (see FIG. 3 and FIG. 6) provided on an outer surface of the wall portion **13** of the body portion **12**. A method of joining the plate-shaped portion **111** and the plate-shaped portion **121** is not limited to welding, and fastening using a bolt or a rivet may be adopted as the joining method.

(48) As described above, in the reinforcing member **100** according to the present embodiment, each of the pair of reinforcing walls **101** disposed opposite to each other is configured by overlapping and joining the plate-shaped portions **111** and **121**. Accordingly, the rigidity of the reinforcing wall **101** is higher than that of the connecting portion **102** that is a single plate-shaped portion. From the viewpoint of thickness, a thickness of the reinforcing wall **101** is larger than a thickness of the connecting portion **102**. With such a configuration, it is possible to enhance effectiveness of a function of suppressing an expansion of the outer case **10** realized by the reinforcing walls **101** while exerting a restraining function by the connecting portion **102** with respect to the two reinforcing walls **101**.

(49) As described above, the energy storage apparatus **1** according to the present embodiment includes the outer case **10** that accommodates the energy storage devices **20** and the reinforcing member **100**. The reinforcing member **100** includes the pair of reinforcing walls **101** disposed so as to sandwich the two wall portions **13** (wall portions **13a**) of the outer case **10** facing each other from the outside, and the connecting portion **102** connecting the pair of reinforcing walls **101** to each other. The rigidity of each of the pair of reinforcing walls **101** is higher than the rigidity of the

connecting portion **102**.

(50) According to this configuration, the pair of reinforcing walls **101** is connected to each other so as to restrict the movement of the pair of reinforcing walls **101** to the outside. With such a configuration, the expansion of the wall portions **13a** to the outside, when the wall portions **13a** of the outer case **10** receives a force (an internal pressure of the outer case **10** or a pressing force attributed to an expansion of one or more energy storage devices **20**) from the inside, can be suppressed. As a result, the occurrence of a drawback such as a damage on the outer case **10** is suppressed. When the valve of the energy storage device **20** in the outer case **10** is opened so that a gas is released from the outer case **10**, an internal pressure of the outer case **10** in a gastight state or in a substantially gastight state is rapidly increased. In this case, there may be a case where the release of the gas from the exhaust tube **90** does not catch up with the increase of the internal pressure, and the internal pressure of the outer case **10** increases or the internal pressure is maintained at a high level. In this case, it is conceivable that a damage such as cracking occurs in the resin-made outer case **10**, and the gas leaks out from the damaged portion. That is, there is a possibility that the gas leaks from an unexpected place of the outer case **10**, and that may cause a problem on safety. However, in the energy storage apparatus **1** according to this embodiment, the expansion of the outer case **10** is suppressed by the reinforcing member **100** fixed to the outer case **10** and hence, a possibility that a damage such as cracking occurs in the outer case **10** is reduced. Such an advantageous effect can be obtained by the reinforcing member **100** that is attached to the outer case **10** from the outside and hence, the above-mentioned advantageous effect can be obtained without changing the designing of the outer case **10**.

(51) Further, the reinforcing wall **101** has the relatively high rigidity and hence, the expansion of the wall portion **13** of the outer case **10** can be more reliably suppressed. Further, with respect to the connecting member **102** where a demand for high rigidity is not so high compared to the reinforcing wall **101**, a thickness of the connecting member **102** can be reduced or the connecting member **102** is thinned. As illustrated in FIG. 3, the connecting member **102** may be light-weighted by forming four thinned portions (opening portions). That is, in the energy storage apparatus **1** of this embodiment, it is possible to effectively suppress the deformation of the outer case **10** (the expansion of the wall portion **13**) while realizing the reduction of weight, downsizing or the like of the reinforcing member **100**. As described above, according to the energy storage apparatus **1** of the present embodiment, the safety of the energy storage apparatus **1** can be enhanced.

(52) The comparison in rigidity between the reinforcing wall **101** and the connecting portion **102** is performed by comparing values in bending rigidity between the reinforcing wall **101** and the connecting portion **102**. The bending rigidity of the reinforcing wall **101** means a bending rigidity in an arrangement direction (X axis direction) of the reinforcing wall **101** and the outer case **10**. The bending rigidity of the connecting portion **102** means a bending rigidity in an arrangement direction (Y axis direction) of the connecting portion **102** and the outer case **10**. Specifically, in the reinforcing member **100**, the pair of reinforcing walls **101** and the connecting portion **102** are cut at the connecting portion, and the pair of reinforcing walls **101** and the connecting portion **102** are separated from each other. Thereafter, the bending rigidity of the pair of reinforcing walls **101** and the bending rigidity of the connecting portion **102** are measured, and measured bending rigidity values are compared with each other. As described in the present embodiment, such a difference in rigidity may be realized by the difference in cross-sectional shape (cross-sectional secondary moment) between the reinforcing wall **101** and the connecting portion **102**, the difference in material between the reinforcing wall **101** and the connecting portion **102**, or the like, besides the difference in thickness between the reinforcing wall **101** and the connecting portion **102**.

(53) In the present embodiment, the pair of reinforcing walls **101** are connected to each other by the connecting portion **102** disposed on both sides in a width direction (Y-axis direction) that intersects with an oppositely facing direction of the two wall portions **13**.

(54) As described above, in the present embodiment, the pair of reinforcing walls **101** is brought

into a posture parallel to the YZ plane respectively and is connected to each other on both sides in the Y axis direction and hence, the pair of reinforcing walls **101** can exert a high restraining force with respect to the outer case **10**. With such a configuration, an effect of suppressing the expansion of the outer case **10** is enhanced, and this contributes to the enhancement of the safety of the energy storage apparatus **1**.

(55) In the present embodiment, as illustrated in FIG. **2**, the plurality of energy storage devices **20** arranged side by side in an opposedly facing direction of the two wall portions **13** (wall portions **13a**) are accommodated in the outer case **10**.

(56) To be more specific, in the outer case **10**, each of the plurality of energy storage devices **20** is a prismatic energy storage device **20**, and is disposed such that the long-side surfaces **21a** (see FIG. **2**) are directed in the arrangement direction of the energy storage devices **20**. In the prismatic energy storage device **20**, the long-side surfaces **21a** are likely to expand and hence, the energy storage device array **24** formed of the plurality of energy storage devices **20** is likely to expand in the arrangement direction of the plurality of energy storage devices **20**. Accordingly, it can be said that the two opposedly facing wall portions **13a** are likely to expand outward by receiving the expansion of the energy storage device array **24**. In this regard, in the energy storage apparatus **1** of the present embodiment, the reinforcing member **100** is disposed in a posture suitable for restraining the two wall portions **13a** and hence, the expansion of the outer case **10** can be effectively suppressed. That is, the reinforcing member **100** is disposed so as to press the portion of the outer case **10** that is likely to expand. With such a configuration, the safety of the energy storage apparatus **1** is enhanced.

(57) In the present embodiment, the reinforcing member **100** includes the first reinforcing member **110** and the second reinforcing member **120**. Each of the first reinforcing member **110** and the second reinforcing member **120** includes the two plate-shaped portions **111** or **121** disposed opposite to each other and the connecting portion **102** connecting the two plate-shaped portions **111** or **121** to each other. Each of the pair of reinforcing walls **101** is configured by overlapping one of the two plate-shaped portions **111** of the first reinforcing member **110** and one of the two plate-shaped portions **121** of the second reinforcing member **120**.

(58) According to such a configuration, the reinforcing member **100** can be configured by combining two members (first and second reinforcing members **120**) having the same shape with each other. Accordingly, it is possible to obtain a reinforcing wall **101** having high rigidity that is formed of the two plate-shaped portions **111**, **121**. By combining the two members such that the connecting portions **102** of the two members face each other, it is also possible to obtain the reinforcing member **100** that can suppress the expansion of the outer case **10** in a well-balanced manner. Accordingly, it is possible to obtain the reinforcing member **100** that efficiently suppresses the expansion of the outer case **10** with the simple configuration, and this contributes to the enhancement of the safety of the energy storage apparatus **1**.

(59) The configuration of the energy storage apparatus **1** according to the present embodiment can also be described as follows. That is, the energy storage apparatus **1** includes: the outer case **10** that accommodates the energy storage devices **20**; and the reinforcing member **100** that includes the pair of reinforcing walls **101** and the connecting portions **102**. The pair of reinforcing walls **101** is disposed so as to sandwich the two wall portions **13** (wall portions **13a**) of the outer case **10** opposedly facing each other from the outside. The connecting portion **102** connects the pair of reinforcing walls **101** to each other. The reinforcing member **100** includes the first reinforcing member **110** and the second reinforcing member **120**, and the first reinforcing member **110** includes the two plate-shaped portions **111** disposed opposite to each other and the connecting portion **102** that connects the two plate-shaped portions **111** to each other. The second reinforcing member **120** includes the two plate-shaped portions **121** disposed opposite to each other and the connecting portion **102** that connects the two plate-shaped portions **121** to each other. Each of the pair of reinforcing walls **101** is configured by overlapping one of the two plate-shaped portions **111** of the

first reinforcing member **110** and one of the two plate-shaped portions **121** of the second reinforcing member **120**.

(60) According to this configuration, the pair of reinforcing walls **101** is connected to each other so as to restrict the movement of the pair of reinforcing walls **101** to the outside and hence, the expansion of the wall portion **13a** of the outer case **10** to the outside is suppressed. As a result, the occurrence of a damage or the like on the outer case **10** is suppressed. The reinforcing member **100** can be configured by combining two members (first and second reinforcing members **120**) having the same shape with each other. Accordingly, it is possible to obtain a reinforcing wall **101** having high rigidity that is formed of the two plate-shaped portions **111**, **121**. By combining the first reinforcing member **110** and the second reinforcing member **120** such that the connecting portions **102** of the first and second reinforcing members **110**, **120** face each other, it is also possible to obtain the reinforcing member **100** that can suppress the expansion of the outer case **10** in a well-balanced manner. Accordingly, it is possible to obtain the reinforcing member **100** that efficiently suppresses the expansion of the outer case **10** with a simple configuration. Therefore, according to the energy storage apparatus **1** of the present embodiment, the safety of the energy storage apparatus **1** can be enhanced.

(61) The configuration of the energy storage apparatus **1** according to the present embodiment can also be described as follows. That is, the energy storage apparatus **1** includes the outer case **10** that accommodates the energy storage devices **20** and the reinforcing member **100**. The reinforcing member **100** includes the pair of reinforcing walls **101** disposed so as to sandwich the two wall portions **13** (wall portions **13a**) of the outer case **10** facing each other from the outside, and the connecting portion **102** connecting the pair of reinforcing walls **101** to each other. The thickness of each of the pair of reinforcing walls **101** is larger than the thickness of the connecting portion **102**.

(62) According to this configuration, the pair of reinforcing walls **101** is connected to each other so as to restrict the movement of the pair of reinforcing walls **101** to the outside. Accordingly, when the wall portions **13a** of the outer case **10** receive an external force from the inside, the expansion of the wall portions **13a** to the outside is suppressed. As a result, the occurrence of a drawback such as a damage on the outer case **10** is suppressed. Such an advantageous effect can be obtained by the reinforcing member **100** that is attached to the outer case **10** from the outside and hence, the above-mentioned advantageous effect can be obtained without changing the designing of the outer case **10**. Further, the reinforcing wall **101** has the relatively large thickness and hence, the expansion of the wall portion **13** of the outer case **10** can be more reliably suppressed. Further, with respect to the connecting portion **102** where a demand for large thickness is not so high compared to the reinforcing wall **101**, the connecting portion **102** can be formed with a relatively thin wall thickness. That is, it is possible to effectively suppress the deformation of the outer case **10** (the expansion of the wall portion **13**) while realizing the reduction of weight, downsizing or the like of the reinforcing member **100**. Therefore, according to the energy storage apparatus **1** of the present embodiment, the safety of the energy storage apparatus **1** can be enhanced.

(63) The thickness of the reinforcing wall **101** is a width of the reinforcing wall **101** in the direction in which the pair of reinforcing walls **101** is arranged. The thickness of the reinforcing wall **101** is a width, in the arrangement direction, of main portions excluding protrusions protruding in the arrangement direction, through holes penetrating the reinforcing wall **101** in the arrangement direction, recessed portions recessed in the arrangement direction, and the like. In a case where there exists distribution or unevenness in the thickness of the reinforcing wall **101** as viewed in the direction that the reinforcing wall **101** is arranged, an average value, a most frequent value, or the like of the thickness may be adopted as the thickness of the reinforcing wall **101**.

(64) In the present embodiment, the reinforcing wall **101** thicker than the connecting portion **102** is formed by overlapping the plate-shaped portions **111** and **121**. However, the reinforcing wall **101** may be formed of a single member. A reinforcing member **100** having a reinforcing wall **101** thicker than a connecting portion **102** may be manufactured by using a metal plate in which a thick

portion and a thin portion are formed by press working.

(65) In the present embodiment, more specifically, as shown in FIG. 6, the lid body **11** and the body portion **12** of the outer case **10** are joined to each other at a joint portion **14** positioned at a peripheral edge of the opening portion **15** (see FIG. 2) of the outer case **10**. For example, the joint portion **14** that is a portion where the lid body **11** and the main body portion **12** are joined to each other is formed by thermal welding. In this configuration, when an internal pressure of the outer case **10** increases, the internal pressure acts on the wall portion **13a** in FIG. 6 in a direction that an upper end of the wall portion **13a** falls outward. Accordingly, stress is likely to concentrate on the joint portion **14**, and as a result, there is a possibility that a part of the joint portion **14** is damaged so that a gas filled in the inside of the outer case **10** leaks out from the part of the joint portion **14**. However, in the energy storage apparatus **1** according to this embodiment, as illustrated in FIG. 6, the reinforcing member **100** having the relatively high rigidity is disposed at a position where the reinforcing member **100** presses the wall portion **13a** from the outside. With such a configuration, the deformation of the wall portion **13a** to the outside is suppressed. As a result, a damage such as cracking of the joint portion **14**, that is a portion where a damage due to deformation of the wall portion **13a** to the outside is likely to occur, is suppressed.

(66) Further, in the present embodiment, as illustrated in FIG. 6, the restricting member **200** is disposed outside the reinforcing member **100**. The restricting member **200** includes a contact portion **201** that is brought into contact with the upper surface of the lid body **11** and a fixing portion **202** fixed to the reinforcing member **100**. The fixing portion **202** is joined to the outer surface of the reinforcing wall **101** of the reinforcing member **100** by welding or the like. Accordingly, the restricting member **200** functions as a member that increases the rigidity of the reinforcing wall **101** or a member that increases the thickness of the reinforcing wall **101**. That is, the fixing portion **202** of the restricting member **200** can enhance the reinforcing function by the reinforcing member **100**.

(67) The fixing portion **202** of the restricting member **200** further has a portion that is brought into contact with a peripheral wall portion **11b** of the lid body **11** from the outside. Accordingly, when the combination of the wall portion **13a** that is a part of the body portion **12** and the peripheral wall portion **11b** of the lid body **11** is defined as the wall portion **16** of the outer case **10**, and the restricting member **200** fixed to the reinforcing member **100** by welding or the like is defined as a part of the reinforcing member **100**, the following can be said. The reinforcing member **100** includes the pair of reinforcing walls **101** disposed so as to sandwich the two wall portions **16** of the outer case **10** oppositely facing each other from the outside, and the connecting portion **102** connecting the pair of reinforcing walls **101** to each other. That is, the restricting member **200** that forms a part of the reinforcing member **100** can function as a member that suppresses the deformation, to the outside, of the peripheral wall portion **11b** of the lid body **11** that forms a part of the wall portion **16** of the outer case **10**. To be more specific, the deformation of the peripheral wall portion **11b** of the lid body **11** caused by the deformation of the wall portion **13a** of the body portion **12** to the outside (the lower end of the peripheral wall portion **11b** is inclined to open outward) is suppressed by the restricting member **200** that forms a portion of the reinforcing member **100**. As described above, according to the reinforcing member **100** of the present embodiment, the expansion of the wall portion **16** formed of the body portion **12** and the lid body **11** can be suppressed, and as a result, the safety of the energy storage apparatus **1** is enhanced.

(68) (Modification)

(69) The energy storage apparatus according to the present invention has been described above based on the embodiment. However, the present invention is not limited to the above-mentioned embodiment. Modes obtained by applying various modifications made by those skilled in the art to the above-mentioned embodiments are also included in the scope of the present invention without departing from the gist of the present invention.

(70) The two connecting portions **102** included in the reinforcing member **100** are not necessarily

disposed in an opposedly facing manner. The reinforcing member **100** may have a connecting portion **102** that connects end portions of the pair of reinforcing walls **101** on the Y-axis direction negative side to each other, and a connecting portion **102** that connects end portions of the pair of reinforcing walls **101** on the Z-axis direction negative side to each other. The number of the connecting portions **102** included in the reinforcing member **100** may be 1 or 3 or more. That is, it is sufficient for the connecting portion **102** to connect the pair of reinforcing walls **101** to each other so as to restrain the pair of reinforcing walls **101** disposed opposite to each other in the X axis direction in the X axis direction, and the shape, the number, the arrangement position, and the like of the connecting portion **102** are not particularly limited.

(71) The reinforcing member **100** is not necessarily formed of the combination of the first reinforcing member **110** and the second reinforcing member **120**. After the connecting portion **102** and the reinforcing wall **101** are manufactured as separate members, the connecting portion **102** and the reinforcing wall **101** may be connected to each other by welding, fastening, or the like. That is, provided that the energy storage apparatus **1** includes the pair of reinforcing walls **101** and at least one connecting portion **102** connecting the pair of reinforcing walls **101** to each other, and the rigidity of the reinforcing wall **101** is higher than the rigidity of the connecting portion **102**, the number, the shape, and the like of the members constituting the reinforcing member **100** are not particularly limited. In this supplementary term, “provided that the rigidity of the reinforcing wall **101** is higher than the rigidity of the connecting portion **102**,” may be read as “when the thickness of the reinforcing wall **101** is larger than the thickness of the connecting portion **102**”. In any case, the deformation of the outer case **10** (expansion of the wall portion **13**) can be effectively suppressed by the reinforcing member **100**. Accordingly, the safety of the energy storage apparatus **1** is enhanced.

(72) Each of the reinforcing member **100** and the restricting member **200** is not necessarily made of metal. Each of the reinforcing member **100** and the restricting member **200** may be formed of a non-metallic material having high rigidity such as fiber-reinforced plastic.

(73) The energy storage apparatus **1** may not include the restricting member **200**. The energy storage apparatus **1** includes at least the reinforcing member **100** as the member for reinforcing the outer case **10** and hence, it is possible to enhance the safety of the energy storage apparatus **1** as described above.

(74) The configurations that are formed by arbitrarily combining the above-described plurality of constituent elements also fall within the scope of the present invention.

INDUSTRIAL APPLICABILITY

(75) The present invention is applicable to an energy storage apparatus that includes energy storage devices such as lithium ion secondary batteries or the like.

DESCRIPTION OF REFERENCE SIGNS

(76) **1**: energy storage apparatus **10**: outer case **11**: lid body **11b**: peripheral wall portion **12**: body portion **13**, **13a**, **13b**, **16**: wall portion **13e**: rib **14**: joint portion **15**: opening portion **20**: energy storage device **21**: case **21a**: long-side surface **100**: reinforcing member **101**: reinforcing wall **102**: connecting portion **110**: first reinforcing member **111**, **121**: plate-shaped portion **120**: second reinforcing member **200**, **210**, **220**, **230**: restricting member **201**: contact portion **202**: fixing portion

Claims

1. An energy storage apparatus comprising: an outer case that accommodates an energy storage device; and a reinforcing member including: a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions opposedly facing each other; and a connecting portion that connects the pair of reinforcing walls to each other, wherein rigidity of each of the pair of reinforcing walls is higher than rigidity of the

connecting portion.

2. The energy storage apparatus according to claim 1, wherein the pair of reinforcing walls is connected by the connecting portion disposed on each of both sides in a width direction that intersects with an oppositely facing direction of the two wall portions.

3. The energy storage apparatus according to claim 1, wherein the energy storage device includes a case that includes a pair of long-side surfaces oppositely facing each other, and a plurality of the energy storage devices are accommodated in the outer case, the plurality of the energy storage devices being arranged side by side such that the pair of long-side surfaces faces the oppositely facing direction of the two wall portions.

4. The energy storage apparatus according to claim 1, wherein the reinforcing member includes a first reinforcing member and a second reinforcing member, each of the first reinforcing member and the second reinforcing member includes two plate-shaped portions disposed opposite to each other and the connecting portion connecting the two plate-shaped portions, and each of the pair of reinforcing walls is formed by making one of the two plate-shaped portions of the first reinforcing member and one of the two plate-shaped portions of the second reinforcing member overlap with each other.

5. An energy storage apparatus comprising: an outer case that accommodates an energy storage device; and a reinforcing member including: a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions oppositely facing each other; and a connecting portion that connects the pair of reinforcing walls to each other, wherein the reinforcing member includes a first reinforcing member and a second reinforcing member, each of the first reinforcing member and the second reinforcing member includes two plate-shaped portions disposed opposite to each other and the connecting portion connecting the two plate-shaped portions, and each of the pair of reinforcing walls is formed by making one of the two plate-shaped portions of the first reinforcing member and one of the two plate-shaped portions of the second reinforcing member overlap with each other.

6. An energy storage apparatus comprising: an outer case that accommodates an energy storage device; and a reinforcing member including: a pair of reinforcing walls disposed so as to sandwich two wall portions of the outer case from an outside of the outer case, the two wall portions oppositely facing each other; and a connecting portion that connects the pair of reinforcing walls to each other, wherein a thickness of each of the pair of reinforcing walls is larger than a thickness of the connecting portion.
